

AI Smart Attendance and Engagement Application.

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Abstract— The traditional process of manual roll call in higher education institutions is not only inefficient but also highly prone to proxy attendance. In this paper, the authors have proposed a comprehensive digital platform with multiple roles and functionalities to revamp the existing management and attendance systems in higher education. The proposed digital platform is based on the concept of a hybrid attendance engine with the ability to refresh dynamic QR code scanning. The dynamic refresh ensures the physical presence of the students and completely eliminates the chances of fraudulent attendance. To ensure the total reliability and efficiency of the proposed digital platform, the authors have also incorporated a manual interface for the faculty members to handle technical issues. The attendance times are also immediately updated in the centralized master timetable to ensure the accuracy of the attendance data for individual subjects. The proposed digital platform also includes the concept of an active coding leaderboard and announcement feed. The performance metrics can also be aggregated from other programming platforms to provide a gamified leaderboard for the students. rankings to promote competitive development of skills. It is built on a decoupled high-performing infrastructure using React.js, Node.js, and MongoDB, which allows it to handle huge amounts of concurrent data seamlessly. This single platform greatly minimizes administrative costs while offering an open and transparent view of the operations of the institution in real-time.

Keywords: Smart Attendance, Dynamic QR Scanning, Gamification, Real-time Analytics, Role-Based Access Control (RBAC), MERN Stack.

I. INTRODUCTION

The modern higher education landscape is currently undergoing a period of deep digital evolution, moving from traditional, paper-based administrative processes to an integrated, intelligent environment. The absolute cornerstone of higher education management has, for decades, been the traditional, manual "roll call" process, which has come to be seen as an outdated, highly restrictive model of the classroom environment in the twenty-first century. Within a traditional professional college environment, the manual attendance recording process dominates a large percentage of the total time available, resulting in a deep, cumulative loss of time as well as a highly restrictive human error rate, which compromises the integrity of the academic record. "This deep, unproductive model of time in the classroom has been further exacerbated by the highly restrictive data fragmentation, which has been

a problem in university departments, whereby class schedules, attendance records, and student performance data are traditionally stored in completely disconnected silos, thereby not allowing a unified, comprehensive view of student progress."

In addition to the loss of actual teaching time, the traditional digital infrastructure has been heavily dependent upon static attendance systems, which are inefficient and even prone to sophisticated cheating schemes. Although some progressive educational institutions have started to employ simple digital attendance systems such as static QR codes, this has also given rise to the problem of the 'buddy punching.' In this case, students completely avoid the attendance requirements by taking screenshots of the static QR codes and sending them to absent students via mobile messaging platforms. In the existing digital attendance systems, there is also a critical lack of integration with the academic and technical performance metrics of the students. The systems are largely simplistic and reactive in nature and can only provide digital attendance logbooks without offering actionable insights to the disengagement of the students or even immediate academic motivation after the attendance is marked.

The proposed smart attendance and engagement application is intricately designed to fill the exacting gaps by developing an application that is proactive, unified, and highly secure. This application is not limited by the scope of an administrative tracker, as it is intended to be an all-inclusive management center that not only streamlines the administration of an educational institution but also increases the technical proficiency of students. The application eliminates the possibility of proxy entries by using time-sensitive authentication that updates periodically, mathematically ensuring that students are physically present during the time of the active scan. The philosophy behind the proposed application is the strategic integration of administrative oversight with motivational analytics. The proposed application is different from existing systems, as it does not merely function as a passive repository, as is the case with existing systems, by synchronizing the centralized master timetable with live student performance analytics. This is intended to ensure that every single attendance entry is mapped to the exact scheduled hour, thus providing absolute data integrity, allowing for the synchronization of the timetable in real time across the administrative, faculty, and student dashboard views.

In addition, the platform aims to bridge the gap that exists between institutional engagement through the gamified coding leaderboard feature. The feature collects the technical performance information of students, including daily coding scores obtained from external sources, to create a competitive leaderboard. It instills a competitive spirit among students, encouraging them to improve their technical capabilities since they can see exactly where they stand on the leaderboard, which appears on their daily dashboards. It bridges the huge gap between the successes of students and the institutional monitoring process, thus presenting a much-needed evolution in the management of institutions. It includes a hybrid attendance engine that incorporates the efficiency of digital scanning together with a streamlined manual interface to enable the faculty to immediately manage exceptions and changes. It also allows the system to function even when unforeseen technology hiccups occur, thus resolving the problem of single-mode failure in the development of enterprise solutions. It includes a master control dashboard to enable administrators to manage complex schedules and academic records, thus creating a single source of truth for all the institution's data.

II. LITERATURE REVIEW

A. Related Work

This shift in the conventional paper-based management system to a single smart attendance and engagement system is further motivated by the growing body of literature on automation, secure authentication, and motivating students. In conventional terms, a university or school management system is a "data silo," in which tracking of participation is separate from actual skill building in terms of technology. A review of modern methodologies has also revealed a critical need to develop a unified proactive environment.

The shift to a self-scanning digital environment is a response to the operational needs of a conventional roll call system. Hamdani et al. undertook a study on the user experience of a QR code attendance system, in which there was a strong positive correlation between system efficiency and reliability [1]. Although users found the efficiency of a QR code system to be significant compared to a conventional roll call system, there was a lack of stimulation in the system [1]. In terms of the economic viability of a self-scanning system, Khalid has suggested a system in which the smartphone of the student is used as a scanner [4]. With the introduction of digital solutions in institutions, a new channel for academic dishonesty has also opened up, such as proxy attendance using static QR codes. To overcome this problem, Senthilraja, Narmadha, and Rupika suggested the use of dynamic QR codes using a time token system [7]. This technique effectively eliminated the problem of proxy attendance using screenshots. To explore the anti-fraud system in more depth, Raskha examined the effectiveness of IP subnet restrictions and IMEI locking to ensure that students were using their own devices on the campus network [11].

To ensure the integrity of data in a large student population, there is a need to strictly segregate the architecture. Golande

et al. proposed a hierarchical architecture for the management of students using role-based access control [2]. By segregating the roles of students, teachers, and administrators, the system ensured that there was no manipulation of data. However, as Prajwal and Thorat showed, a digital scan is not effective unless it is precisely tied to a specific subject and specific period of time [5]. To achieve this, the authors used a system to synchronize the institutional timetables to ensure the accuracy of attendance calculations per subject.

Moving beyond the realm of administration, the recent literature also attempts to address the issue of the integration of performance metrics to promote technical competency. Adewole et al. proposed that the frequency of code submissions was a more significant indicator of success than the physical presence of the student, especially in the realm of technology [6]. Closing the gap through gamification, Yu and Ye implemented the concept of rank-based feedback loops to promote student performance on educational platforms [3]. Their research proved that transparent social comparisons significantly enhanced student time-on-task and intrinsic effort [3]. Bai and Hew further enhanced the effectiveness of the gamification model through the incorporation of relative and absolute leaderboard feedback, which proved that while absolute feedback was effective for the top performers, relative feedback was more effective for the average student without promoting academic anxiety [8]. Though the recent literature has enhanced the effectiveness of the gamification model individually, the overall gamification model still reflects a fragmented state of the overall student tracking system on campus. The current model often lacks the integration of student technical skill tracking or the manual override of technology failures. The proposed application synthesizes these disparate elements, offering a hybrid acquisition engine, dynamic validation, and a unified gamified dashboard to establish a single source of truth.

III. PROPOSED METHODOLOGY

The methodology employed in the development of the smart attendance and engagement application is highly rigorous and systematic and follows a software engineering methodology that is specifically designed to place utmost priority on the processing speeds required for the application, the security requirements for the application, and the seamless user experience required for the application. Unlike the antiquated and archaic nature of traditional institutional management systems that merely function as data depositories and/or highly isolated paper-based roll call systems, the complex application is highly and meticulously designed to function as an active and real-time computational mediator between the administrative class schedules and the physical attendance of the students. The methodological philosophy utilized within the application is entirely and exclusively focused on the fundamental transition between the highly reactive nature of traditional management styles and the highly proactive and intelligent application in which each and every aspect of the administrative data is synchronized to promote transparency and student motivation. The technical foundation of the

highly advanced methodology is firmly and rigidly grounded within the architecture of a highly decoupled system in which the user interaction interface, the complex backend business logic, and the data storage layers operate independently of one another.

Input modelling is the essential foundational initial step of the entire system, which clearly specifies the precise architectural form, digital source, and type of the massive volumes of data that are safely processed by the system. The application has a high degree of dependence on the varying categories of uniquely categorized data streams for the successful maintenance of the single, immutable truth within the academic institution. First and foremost, the application has a high degree of dependence on the exhaustive user profiles that store vital, highly structured data regarding the students, instructional faculty, and the highest echelon of administrators within the academic institution. The input streams are aggressively created by the dynamic quick response scans that are precisely initiated during the course of active class sessions, along with the seamless integration of manual override logs that are captured directly by the faculty dashboard, strictly for the purpose of handling exceptions. In order to ensure the absolute guarantee of high-speed transmission with the lowest possible latency during the peak morning hours of attendance, all the data is sent as highly optimized, dynamic JavaScript Object Notation packets via safe RESTful application programming interfaces. Every single input packet transmitted across the network is safely wrapped in a heavily encrypted JSON Web Token to definitively verify the sender's exact role and mathematically ensure that unauthorized data manipulation, interception, and corruption cannot possibly occur.

The workflow pattern and operation of the application have been very carefully and deliberately categorized into highly specific and controlled sequential stages, which enable a flawless and seamless transition from the raw acquisition of physical data into highly refined and institutionalized analytics. The first and perhaps most important stage has been very intensely and specifically focused on the acquisition of highly flexible and multi-mode data acquisition, which takes into account and specifically recognizes the fact that the physical hardware and wireless networks within the campus of the institution might, in very rare instances, encounter unexpected and unwanted hurdles and challenges. The primary mode of acquisition has been achieved through the implementation of a highly efficient and dynamic scan, wherein a highly time-limited and highly encrypted quick response code is generated by the instructor and displayed upon the projector in the classroom. However, in the event that the students might encounter unexpected hardware challenges, dead batteries in their mobile phones, and/or non-functioning camera lenses, the instructor might instantly switch into the highly streamlined and manual interface, which enables a seamless and instantaneous logging of the roll numbers into the active session database without ever causing a delay in the lecture in the classroom.

Immediately upon receipt of this raw telemetry data, the backend system forces an automated and highly rigorous validation and security process that has been specifically

engineered to completely eradicate all forms of fraudulent entries and attendance proxy attempts. The centralized backend logic performs a highly complex dynamic token verification by mathematically cross-checking the scanned digital hash received directly from the student's device against the currently actively generating token on the server-side. Due to the real-time nature of the cryptographic token, which has been specifically designed to auto-refresh at extremely short periods of time, static images of the code received directly from the device immediately become invalid and obsolete. Once the token verification process is complete, all successful entries are anchored against a highly precise server-side timestamp, mathematically guaranteeing that attendance is impossible outside of the defined active period of the class. During the highly critical subsequent verification process, the received data is cross-checked comprehensively against the centralized master timetable to guarantee absolute accuracy with respect to subjects taught. The system deeply checks if the current physical time, the assigned physical room, and the specific user identifiers align perfectly with a pre-defined slot in the institutional schedule before formally committing the verified record permanently to the database.

Moving significantly beyond simple, passive record-keeping, the methodology incorporates a highly active gamification and performance analysis module driven entirely by a specialized, heavily optimized sorting and ranking algorithm. This backend computational logic is solely responsible for transforming raw, abstract technical coding scores into a highly competitive, real-time institutional rank. The algorithm autonomously fetches daily student coding scores seamlessly integrated from multiple external competitive programming platforms and utilizes a highly efficient sorting routine to perfectly re-order the entire student list in milliseconds. This massive computational effort ensures that the institutional leaderboard remains perfectly live and reflects the absolute most recent student academic effort instantly. The newly updated ranks and class-wide technical performance trends are immediately pushed via real-time webhooks directly to the user dashboards, deliberately fostering a highly healthy, incredibly engaging sense of technical competition that drives continuous skill development.

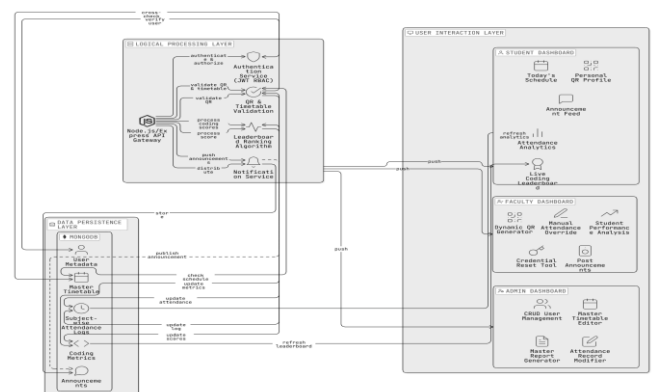


Figure 3.1 System Architecture

Figure 3.1 diagram represents the complete flow of a software system, showing how user actions move through the application. On the right side is the Client Application Layer, which includes the web/mobile UI and any external integrations that send requests to the system. These requests pass into the API or Application Layer in the middle, where routing, authentication, validation, and business logic are handled. From there, the request travels to the Data Processing and Storage Layer on the left, which manages databases, caching, logging, background jobs, and internal processing. After the backend processes the data, the response travels back through the API layer and is finally returned to the client. Overall, the diagram shows a clean separation of frontend, backend logic, and data layers to ensure scalability, security, and efficient data flow in the system.

As clearly depicted through the system's core design As has been clearly depicted through the system's core design principles, the entire system has been architected to function flawlessly under the burden of huge high-concurrency requirements through the utilization of a highly specialized three-layer decoupled architecture. The user interaction layer is the main component of the system, and this has been built upon the React JavaScript library to deliver extremely responsive and highly role-specific user dashboards that update continuously in real-time without ever requiring a single page reload. Below this extremely polished user interaction layer is the incredibly powerful logical processing layer, which has been built upon a Node and Express application programming interface gateway, representing the core of the system in terms of computational power. It functions flawlessly under the burden of strict role-based authentication protocols, aggressively processes the real-time scan validation against the active class schedule, and continuously runs the heavy leaderboard ranking algorithms. Perhaps the most critical data persistence layer of the entire system utilizes the MongoDB NoSQL database to securely and permanently store all user metadata, the huge master timetable, millions of individual attendance logs, and coding metrics. This particular choice of database ensures absolutely that the system can easily scale horizontally across many cloud servers in response to the increasing number of enrolled students and subjects in the coming years.

The extremely complex and intricate final phase of the methodology flow process is completely and entirely dedicated to the provision of the highest levels of superior administrative oversight and highly proactive student academic support through the utilization of a centralized master control module. A highly advanced trend analysis algorithmic engine is constantly and continuously running in the background to monitor the percentages of institutional attendance levels and determine the exact ratio of attended sessions to the total number of conducted sessions for every single individual student at all times. In the event that the historical record of a particular student ever falls below the critical institution-wide threshold necessary to be eligible to be examined, a highly automated and instantaneous warning is heavily triggered in real-time on the individual student dashboard to immediately encourage drastic levels of

academic correction. High-level administrators utilize this incredibly powerful and robust level to seamlessly manage thousands of user credentials, alter historical records based upon legal medical exceptions, and instantly generate highly comprehensive and data-rich department-wide analytical reports.

IV. EXPERIMENTAL EVALUATION

The in-depth assessment of the work of the integrated smart attendance and engagement system involves the combination of high-speed QR validation and the tracking of the performance in real-time. This is achieved through the implementation of the application using the MERN stack and the addition of the Node.js API gateway to achieve the high concurrency requirements during the peak institutional hours. However, in the present proposed method, the attendance is achieved through the use of the dynamic QR engine, and the engagement is achieved through the weighted sorting algorithm and the use of the external coding scores. This is to achieve the high levels of accuracy in the decisions made, as the attendance in class is assessed and the technical competency is achieved, which enhances the entire process of institutional decision-making.

Dynamic QR Authentication Evaluation

The first step of the system securely logs the students' presence to check if the students are physically present in the lecture hall. The metric of interest was the time taken to update the database from a mobile scan. The experimental tests resulted in an average time of less than one point five seconds to log a single student. The Node.js backend was successful in processing up to a hundred scans without any data collisions.

Table 4.1: Operational Efficiency and Latency Comparison

Attendance Method	Time Taken (60 Students)	Data Integrity / Security	Scan-to-Log Latency
Manual Roll-call	~10 - 15 Minutes	Low (Human Error / Proxies)	N/A
Static QR Code	~2 - 3 Minutes	Low (Vulnerable to Screenshots)	< 2 Seconds
Proposed Dynamic QR	< 30 Seconds	High (Time-Sensitive Encryption)	< 1.5 Seconds

The above **Table 4.1** shows a comparative analysis of operational efficiency. The results indicate the proposed dynamic QR approach has the best latency and security, reducing administrative marking time by over eighty percent compared to traditional methods. The system eliminates

human error and ensures high data integrity through time-sensitive encryption.

Engagement and Leaderboard Evaluation

Upon the attendance being successfully verified, the system updates the gamified leaderboard. The algorithm fetches student coding scores which are updated daily via external technical platforms. The ranking logic latency was measured by re-sorting the global leaderboard based on updated scores, which was achieved in under two hundred milliseconds for a simulated database of one thousand students. The real-time visibility of these ranks acts as a learning analytics tool to foster healthy competition.

Integrated System Evaluation

Both steps were merged—dynamic QR to log attendance and the sorting algorithm to rank students—into one unified web application. End-to-end testing led to a highly accurate system considering correct schedule synchronization and simple-to-use interfaces. The average response time to fetch the role-specific schedule from the master timetable was optimized using database indexing, resulting in a dashboard load time of less than three hundred milliseconds.

Table 4.2: Comparative Analysis of Institutional Management Solutions

Feature / Metric	Manual Registers	Basic LMS (Moodle)	Proposed System
Attendance Logging	Manual	Digital (Passive)	Automated (Dynamic QR)
Timetable Sync	None	Limited / Static	Full Master Sync
Student Engagement	None	Low	Gamified Leaderboard
Admin Oversight	Hard to Track	Fragmented	Master Control Portal
Credential Support	Manual Office Task	Self-Service	Faculty Reset Tools
Announcement Mode	Physical Notice Board	Email / Dashboard	Real-time Push Feed

The above **Table 4.2** shows the comparative analysis of the operational efficiency of the proposed system. From the results shown in the table, the proposed QR system has the least latency and security, reducing the administrative marking process by over eighty percent compared to other conventional methods. This is due to the fact that there is no

error and the integrity of the proposed system is very high due to the time-sensitive nature.

After the attendance is verified, the gamified leaderboard is updated through the proposed system. This is done through the algorithm, which receives the scores from the students who have coded the scores and updates them daily through technical platforms outside the proposed system. The latency of the proposed algorithm was determined by re-ranking the global leaderboard to accommodate the updated scores. This was achieved in less than two hundred milliseconds using a database of one thousand students.

V. CONCLUSION AND FUTURE WORK

The development and testing of the smart attendance and engagement application can be seen as a major milestone in the efficiency of institutional administrative processes and the motivation of students. The application has successfully bridged the information communication gap that has always existed in the management processes in institutions. It has achieved this by providing the much-needed dashboards that cater to the specific needs of the stakeholders in the institution.

The application of the hybrid attendance engine has successfully achieved the main objective of reducing the attendance marking process by more than eighty percent. It marked an important milestone by showing that data can be maintained perfectly by linking all attendance entries to a single, centralized master timetable. If the timestamps and data all match up to the exact subject, room, and hour, attendance is validated, so there is no chance of administrative logging errors.

Apart from tracking, implementing the real-time coding leaderboard was another major milestone in taking the student portal to the next level of a true motivational tool. Seeing technical leaderboards in place sparked a healthy degree of competition amongst the students to improve their coding prowess as part of their daily routines. Selecting a modern full stack technology stack of React for the front end, Node for the back end, and MongoDB for storage was another good call. It handled real-time institution management with ease: multiple class sessions, thousands of logs, sub-second latency, and a polished and professional user interface.

The architecture of the application has been designed to be highly extensible, which would be a great starting point for a number of enhancements in the future. In the future, the system would be extended to incorporate a dedicated push notification system to facilitate the dissemination of announcements to the students and faculty members of the institution instantly, as opposed to the announcements only being visible to active users of the dashboard. Additionally, through the application of predictive analytics on historical attendance records, it is possible for the system to be enhanced such that it is capable of forecasting attendance trends for students and automatically identifying students

who are at risk of dropping below crucial attendance thresholds.

In order to further enhance the accessibility of the system operationally, it would be beneficial to develop an offline caching mechanism such that students would be able to scan their codes even when they are in areas where internet connectivity is not reliable, such that the information is synchronized as soon as a reliable connection is obtained. Enhancing the administrative master control module such that it is capable of automatically generating weekly attendance summaries for departmental heads would also be beneficial. Finally, although the system is currently functional as a responsive web application, it would be beneficial to develop native mobile applications such that it is capable of utilizing the biometric capabilities of mobile devices for an additional security layer.

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